CD705:
Statistical Data Analysis with R

Lecture 6.1: Statistical Inference

Alexey Chernov
A.Chernov@brighton.ac.uk

University of Brighton

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Lecture Outline

1. Statistical Inference
2. Point Estimates
3. Hypothesis Testing
4. Student's *t* Distribution
5. More on Tests and Confidence Intervals
6. Comparing Samples

Statistical Inference

- We want to know some value
- The value cannot be measured precisely for various reasons
  - too expensive (in time, money, people,. . .)
  - unreliable (mixed with random noise, intentionally modified,. . .)
  - unobservable (future value, theoretical construction,. . .)
  - etc.

- We define a random variable related to this value
- We assume a class of possible distributions for the RV
- We obtain some data / observations of the RV
- Based on the data, we restrict the class of possible distributions
  - point estimates
  - interval estimates = confidence intervals
  - hypothesis testing

Cereal Example: Estimates

A machine fills a cereal box to an unknown preset weight between 200 and 500 grams; the standard deviation from the preset weight is 5 grams. A sample of 30 boxes was selected and the average weight of their contents turned out to be 322.4 grams.

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- Let $X_1, \ldots, X_{30}$ be RVs denoting weights of the boxes in the sample and μ be the preset weight
- $\bar{X} = (X_1 + \cdots + X_{30})/30$
- $\mathbf{E}\,\bar{X} = \mu$, hence $\bar{x} = 322.4$ is a point estimate for μ

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- $\bar{X} = (X_1 + \cdots + X_{30})/30$
- $\mathbf{E}\,\bar{X} = \mu$, hence $\bar{x} = 322.4$ is a point estimate for μ
- Due to the CLT, $\bar{X} \sim \mathcal{N}\left(\mu, \frac{5^2}{30}\right)$ approximately, hence 95% CI is

$$\left[322.4 - z^*_{0.975}\frac{5}{\sqrt{30}}, 322.4 + z^*_{0.975}\frac{5}{\sqrt{30}}\right] \approx [320.61, 324.19]$$

where we used $\alpha = 1 - 0.95 = 0.05$ and $z^*_{0.975} \approx 1.95996$

Cereal Example: Hypothesis Testing

The preset weight for the machine was to be set to 325 grams. Does the fact that the average weight of the contents of 30 boxes turned out to be 322.4 grams provide sufficient evidence that the machine was not correctly set?

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The preset weight for the machine was to be set to 325 grams. Does the fact that the average weight of the contents of 30 boxes turned out to be 322.4 grams provide sufficient evidence that the machine was not correctly set?

- Hypothesis: $\mu = 325$, hence $\bar{X} \sim \mathcal{N}\left(325, \frac{5^2}{30}\right)$
- We will calculate $\Pr[\bar{X} \leq 322.4 \text{ or } \bar{X} \geq 327.6]$
- The latter probability is $0.004 = 0.4\%$ is the *p*-value
  - Calculations are in the R Companion
- As the *p*-value is small, we reject the null hypothesis
- Conclusion: the settings of the machine do not look correct

Proportion Example (cf. Week 3)

- We want to know the percentage of “immune people” in the UK
- We select a sample of size n by SRS from the UK population
- The number of “immune people” in the sample is $X \sim B(n, p)$

Proportion Example (cf. Week 3)

- We want to know the percentage of "immune people" in the UK
- We select a sample of size n by SRS from the UK population
- The number of "immune people" in the sample is $X \sim B(n, p)$

- Approximately, $X \sim \mathcal{N}(np, np(1-p))$; hence $\frac{X-np}{\sqrt{np(1-p)}} \sim \mathcal{N}(0,1)$
- Solving $\left|\frac{X-np}{\sqrt{np(1-p)}}\right| \leq z^*_{1-\frac{\alpha}{2}}$ for p, we get an approximate $(1-\alpha)$ confidence interval for p
- $\Pr\left[Z > \left|\frac{x-np_0}{\sqrt{np_0(1-p_0)}}\right|\right]$ is the p-value for the hypothesis $p = p_0$

- See the solution to Exercise 3, Week 3, for an example

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Point Estimates: Some General Ideas

- X is a random variable
- $\mathbf{E}\,X$ is the parameter to be estimated
- A value of X is an estimate for $\mathbf{E}\,X$
- $\mathrm{Var}(X)$ shows how values of X are spread around $\mathbf{E}\,X$
  For small $\mathrm{Var}(X)$, a randomly selected value of X is highly likely to be close to $\mathbf{E}\,X$

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  For small $\mathrm{Var}(X)$, a randomly selected value of X is highly likely to be close to $\mathbf{E}\,X$
- What if $\mathrm{Var}(X)$ is not small? We need to construct another variable Y, such that $\mathbf{E}\,Y = \mathbf{E}\,X$ and $\mathrm{Var}(Y)$ is small!
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- A more advanced way to assess the quality of estimates:
  $\Pr[|X - \mathbf{E}\,X| > \varepsilon]$ is small for some (or many, or all) values of ε;
  ε is called the margin of error
- All distributions of X such that $\Pr[|X - \mathbf{E}\,X| > \varepsilon]$ is small form a confidence interval

Point Estimates: Terminology

Theory	Example
$X_1, \ldots, X_n$ are some RVs and $x_1, \ldots, x_n$ are their observed values	$X_1, \ldots, X_n$ are i.i.d.
Any function of observations $\xi(x_1, \ldots, x_n)$ is called a statistic	$\bar{x} = \dfrac{x_1 + \cdots + x_n}{n}$
ζ is a parameter of the joint distribution of $X_1, \ldots, X_n$	$\mu = \mathbf{E}\, X_1 = \ldots = \mathbf{E}\, X_n$
If we use $\xi(X_1, \ldots, X_n)$ as a "proxy" for ζ, then $\xi(x_1, \ldots, x_n)$ is a point estimate of ζ We say that ξ is an estimator for ζ and write $\hat{\zeta} = \xi(x_1, \ldots, x_n)$	$\hat{\mu} = \bar{x}$ is an estimate $\bar{X} = \dfrac{X_1 + \cdots + X_n}{n}$ is an estimator for μ
$\xi(x_1, \ldots, x_n)$ is an unbiased estimator of ζ if $\mathbf{E}\, \xi(X_1, \ldots, X_n) = \zeta$	$\bar{x}$ is an unbiased estimator of μ

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Lady Tasting Tea: Statement

A lady declares that by tasting a cup of tea made with milk she can discriminate whether the milk or the tea infusion was first added to the cup. [. . .]
Our experiment consists in mixing eight cups of tea, four in one way and four in the other, and presenting them to the subject in a random order. The subject has been told in advance of what the test will consist, namely, that she will be asked to taste eight cups, that these shall be four of each kind, and that they shall be presented to her in a random order, that is in an order not determined arbitrarily by human choice, but by the actual manipulation of the physical apparatus used in games of chance, cards, dice, roulettes, etc. [. . .] Her task is to divide the 8 cups into two sets of 4, agreeing, if possible, with the treatments received.

R. Fisher. *The principles of experimentation, illustrated by a psycho-physical experiment*. [1935]

Lady Testing Tea: Probabilities

If the lady cannot discriminate the cups, then the number of mistakes among, say, tea-first cups follows the hypergeometric distribution:

$$\Pr[\,\text{all correct}\,] = \frac{\binom{4}{0}\binom{8-4}{4-0}}{\binom{8}{4}} = \frac{1}{\frac{8!}{4!4!}} = \frac{1}{70} \approx 0.014$$

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$$\frac{\binom{4}{1}\binom{8-4}{4-1}}{\binom{8}{4}} \approx 0.229$$

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For an experiment with 6 cups each way, the probability of at most one mistake is

$$\frac{\binom{6}{0}\binom{12-6}{6-0}}{\binom{12}{6}} + \frac{\binom{6}{1}\binom{12-6}{6-1}}{\binom{12}{6}} \approx 0.040$$

Some simulations are in the R Companion

Hypothesis Testing: Summary

- Test statistic: a value that can be computed from observations
- The (null) hypothesis: a statement about the random experiment that allows us to calculate the distribution of the test statistic
- "Neighbourhoods" of a test statistic: for each value of the test statistic, we will consider some "similar" values (usually, the more extreme values than the actually observed one)
- *p*-value: the probability of the "neighbourhood" of the actually observed value of the test statistic
- We reject the null hypothesis if the *p*-value is too small
- Significance level
  - *p*-value is too small if it is below the significance level
- Type I error: rejecting the null hypothesis while it is true
- Type II error: not rejecting the null hypothesis while it is not true

Confidence Intervals and Hypothesis Tests

For a given value of a statistic ξ, a $(1-\alpha)$ confidence interval consists of all values ζ_0 of the parameter ζ for which the null hypothesis $\zeta = \zeta_0$ is not rejected at significance level α

This holds under a range conditions.

In the Cereal Example:

- 322.4 is the value of the statistic called the sample mean
- the respective 95% confidence interval is $[320.61, 324.19]$
- this interval does not contain 325
- hence, the hypothesis $\mu = 325$ is rejected at significance level 5%

Speed Cameras Example: The Problem

The transport police in a certain area regard a road suitable for installing a speed camera if the mean speed is at least 35 m.p.h.
At a particular location, a random sample of 24 vehicles shows a mean speed of 36.2 m.p.h. with a standard deviation of 2.7 m.p.h.
Is this location suitable for the installation of a speed camera?

- We need to decide whether the mean speed is greater than a certain threshold
- The sample mean has an approximately normal distribution, where both the mean and the variance are unknown
- The unknown mean is the base for our conclusion
- The unknown variance is necessary only in calculations

One-Sided Confidence Intervals and Tests

- A symmetric $(1-\alpha)$ confidence interval

$$\left[\bar{x} - z^*_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z^*_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$$

  corresponds to testing a "two-sided" hypothesis $\mu = \mu_0$
- We can consider the upper CI (upper confidence bound):

$$[-\infty, \bar{x} + z^*_{1-\alpha} \frac{\sigma}{\sqrt{n}}]$$

  and the lower CI (lower confidence bound):

$$[\bar{x} - z^*_{1-\alpha} \frac{\sigma}{\sqrt{n}}, +\infty]$$

- They correspond to "one-sided" hypotheses $\mu \geq \mu_0$ and $\mu \leq \mu_0$, respectively

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Sample Variance

- When the population variance σ^2 is unknown, we can estimate it from the sample used for estimating the population mean
- Let $x_1, \ldots, x_n$ be a sample from the population
- $\bar{x}$ is an unbiased estimate for the population mean
- The sample variance is

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- It is an unbiased estimate for the population variance
- The sample standard deviation is

$$s_x = \sqrt{s_x^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

- It is NOT an unbiased estimate for the population standard deviation, but we will be using it instead of σ anyway

“Standardising” Using the Sample Standard Deviation

- If $\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$, then $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$
- Let’s plug in the sample standard deviation s_x instead of the unknown σ
- Note that s_x depends on the sample, that is, it is a value of the random variable $S_X = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$
- Hence we need to consider the distribution of

$$\frac{\bar{X}-\mu}{S_X/\sqrt{n}}$$

where $X_1, \ldots, X_n$ are i.i.d. RVs

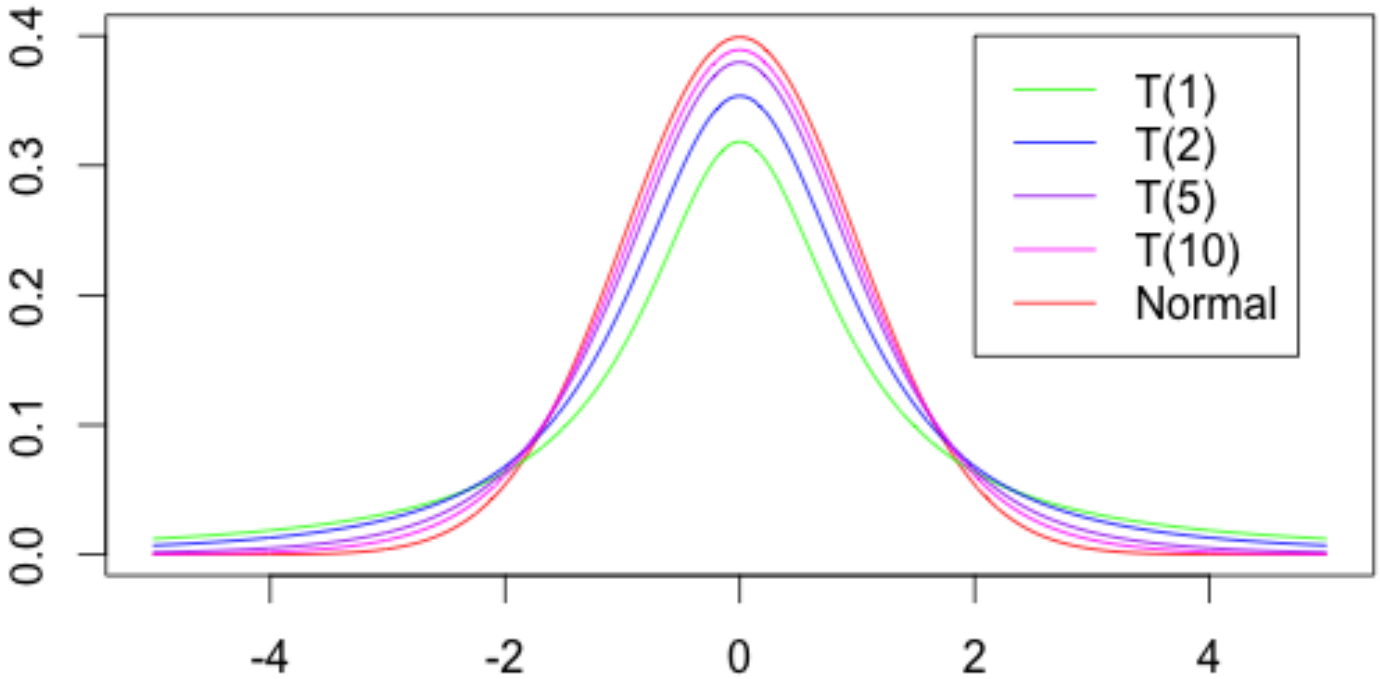
- The distribution of the latter is known if X_i are i.i.d. normal random variables

Student's t-distribution

If $X_1, \ldots, X_n$ are i.i.d. and $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\frac{\bar{X} - \mu}{S_X/\sqrt{n}} \sim T(n-1)$$

$T(n-1)$ is the Student's t-distribution with $(n-1)$ degrees of freedom



Confidence Intervals and Tests Using t-distribution

- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$, $\bar{x}$ be the observed value of $\bar{X}$ and s_x is the observed value of the sample standard deviation
- A $(1-\alpha)$ confidence interval for μ is

$$\left[\bar{x} - t^*_{1-\frac{\alpha}{2},n-1}\frac{s_x}{\sqrt{n}}, \bar{x} + t^*_{1-\frac{\alpha}{2},n-1}\frac{s_x}{\sqrt{n}}\right]$$

- Here $t^*_{1-\frac{\alpha}{2},n-1}$ is the $\left(1-\frac{\alpha}{2}\right)$-quantile of the t-distribution with $(n-1)$ degrees of freedom: $\Pr[T > t^*_{1-\frac{\alpha}{2},n-1}] = \frac{\alpha}{2}$, $T \sim T(n-1)$

Confidence Intervals and Tests Using t-distribution

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- Here $t^*_{1-\frac{\alpha}{2},n-1}$ is the $\left(1-\frac{\alpha}{2}\right)$-quantile of the t-distribution with $(n-1)$ degrees of freedom: $\Pr[T > t^*_{1-\frac{\alpha}{2},n-1}] = \frac{\alpha}{2}$, $T \sim T(n-1)$
- We can also test a hypothesis $\mu = \mu_0$, using $\Pr\left[T > \left|\frac{\bar{x}-\mu_0}{s_x/\sqrt{n}}\right|\right]$ for $T \sim T(n-1)$ as the p-value
- One-sided confidence intervals and tests are defined similarly to the case of known variance

Speed Cameras Example: Two-Sided Test

- The null hypothesis:
  1. the speed of any single car, X, follows the normal distribution: $X \sim \mathcal{N}(\mu, \sigma^2)$, σ^2 is unknown
  2. $\mu \leq 35$
- The test statistic: $\dfrac{\bar{x} - \mu}{s_x/\sqrt{n}} = \dfrac{36.2 - \mu}{2.7/\sqrt{24}}$
- The "neighbourhood" of the test statistic: $\left\{ \dfrac{\bar{X} - \mu}{S_x/\sqrt{24}} \geq \dfrac{36.2 - \mu}{2.7/\sqrt{24}} \right\}$
- If $\mu = 35$, then $\dfrac{\bar{X} - 35}{S_x/\sqrt{24}} \sim T(23)$ and the p-value is

$$\Pr\left[\frac{\bar{X} - 35}{S_x/\sqrt{24}} \geq \frac{36.2 - 35}{2.7/\sqrt{24}} \right] \approx 0.02$$

- Conclusion: reject $X \sim \mathcal{N}(35, \sigma^2)$

Speed Cameras Example: One-Sided Test and CI

- Now assume that $\mu = \mu_0$ and $\mu_0 \leq 35$
- Hence $\dfrac{36.2 - \mu_0}{2.7/\sqrt{24}} \geq \dfrac{36.2 - 35}{2.7/\sqrt{24}}$ and the p-value for $\mu = \mu_0$ is

$$\Pr\left[\frac{\bar{X} - \mu_0}{S_x/\sqrt{24}} \geq \frac{36.2 - \mu_0}{2.7/\sqrt{24}}\right] \leq \Pr\left[\frac{\bar{X} - 35}{S_x/\sqrt{24}} \geq \frac{36.2 - 35}{2.7/\sqrt{24}}\right] \approx 0.02$$

- Hence, we have rejected $X \sim \mathcal{N}(\mu, \sigma^2)$ for all $\mu \leq 35$

Speed Cameras Example: One-Sided Test and CI

- Now assume that $\mu = \mu_0$ and $\mu_0 \leq 35$
- Hence $\dfrac{36.2 - \mu_0}{2.7/\sqrt{24}} \geq \dfrac{36.2 - 35}{2.7/\sqrt{24}}$ and the *p*-value for $\mu = \mu_0$ is

$$\Pr\left[\frac{\bar{X} - \mu_0}{S_x/\sqrt{24}} \geq \frac{36.2 - \mu_0}{2.7/\sqrt{24}}\right] \leq \Pr\left[\frac{\bar{X} - 35}{S_x/\sqrt{24}} \geq \frac{36.2 - 35}{2.7/\sqrt{24}}\right] \approx 0.02$$

- Hence, we have rejected $X \sim \mathcal{N}(\mu, \sigma^2)$ for all $\mu \leq 35$
- The lower confidence bound for μ is

$$\bar{x} - t^*_{1-\alpha,23}\frac{\sigma}{\sqrt{n}} \approx 36.2 - 1.713872 \times \frac{2.7}{\sqrt{24}} \approx 35.25543$$

- We can be 95% confident that $\mu \geq 35.25$

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Null Hypothesis and Assumptions

- Typically, a null hypothesis includes two components:
  1. the family / shape of some distribution
     - ⋆ for example, a normal distribution of each RV
  2. the value(s) of the parameter(s)
- When we reject the null hypothesis, we conclude that the combination of these components does not hold
- However, we cannot know which of the components fails

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- However, we cannot know which of the components fails

- Quite often, the first component is called the assumption(s) and the second component is called the null hypothesis
- This way, we implicitly reformulate our conclusion as "if the assumptions hold, then the value of the parameter is not . . . "

- The assumptions are even more important for CIs as they are always present and cannot be incorporated into the interval

Alternative Hypothesis

Alongside the null hypothesis, we may state an alternative hypothesis

- The alternative hypothesis is something we accept when we reject the null hypothesis
- The alternative hypothesis informs the choice of “neighbourhoods”
- Assumptions are normally the common part of the null and alternative hypotheses

Alternative Hypothesis: Example

- The example about Cereal Boxes uses a two-tailed z-test
  - the assumption is $\bar{X} \sim \mathcal{N}(\mu, 5^2/30)$
  - the null hypothesis is $\mu = 325$
  - the alternative hypothesis is $\mu \neq 325$

Alternative Hypothesis: Example

- The example about Cereal Boxes uses a two-tailed z-test
  - the assumption is $\bar{X} \sim \mathcal{N}(\mu, 5^2/30)$
  - the null hypothesis is $\mu = 325$
  - the alternative hypothesis is $\mu \neq 325$

- The example about Speed Cameras uses a one-tailed t-test
  - the assumption is $X_i \sim \mathcal{N}(\mu, \sigma^2)$ for all i
  - the null hypothesis is $\mu = 35$ (or $\mu \leq 35$)
  - the alternative hypothesis is $\mu > 35$

Type I and Type II error and the Power of the Test

- Type I error: rejecting the null hypothesis while it is true
- Type II error: not rejecting the null hypothesis while it is not true

Type I and Type II error and the Power of the Test

- Type I error: rejecting the null hypothesis while it is true
- Type II error: not rejecting the null hypothesis while it is not true
- The probability of Type I error is not greater than the significance level α
- The alternative hypothesis may allow us to calculate the probability β of Type II error
- The power of the test $1 - \beta$ is the probability of rejecting the null hypothesis while the alternative is true

- Let's look at a simulation in R

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A TV Study Example

A study aims to compare how long males and females spend watching TV. 20 males and 20 females are randomly selected and the TV watching times for them are recorded.

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A study aims to compare how long males and females spend watching TV. 20 males and 20 females are randomly selected and the TV watching times for them are recorded.

- We need to decide whether the mean time for males is equal to the mean time for females are equal
- Now we compare two unknown means
- Again, we assume that both sample means have approximately normal distributions

Two Sample Tests

- The null hypothesis:
  1. the TV watching time a male, $X^{(1)}$, follows the normal distribution: $X^{(1)} \sim \mathcal{N}(\mu_1, \sigma_1^2)$
  2. the TV watching time a female, $X^{(2)}$, follows the normal distribution: $X^{(2)} \sim \mathcal{N}(\mu_2, \sigma_2^2)$
  3. $\mu_1 = \mu_2$

Two Sample Tests

- The null hypothesis:
  1. the TV watching time a male, $X^{(1)}$, follows the normal distribution: $X^{(1)} \sim \mathcal{N}(\mu_1, \sigma_1^2)$
  2. the TV watching time a female, $X^{(2)}$, follows the normal distribution: $X^{(2)} \sim \mathcal{N}(\mu_2, \sigma_2^2)$
  3. $\mu_1 = \mu_2$

- If σ_1 and σ_2 are known, then assuming $\mu_1 = \mu_2$

$$\overline{X^{(1)}} - \overline{X^{(2)}} \sim \mathcal{N}\left(0, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

  (we have n_1 observations for $X^{(1)}$ and n_2 observations for $X^{(2)}$)
- We can use a z-test

Independent Samples t-Tests

- Let σ_1 and σ_2 be unknown

- We may additionally assume that $\sigma_1 = \sigma_2$
- This leads to a two-sample pooled variance t-test:
- The test statistic is

$$\tilde{t} = \frac{\overline{x^{(1)}} - \overline{x^{(2)}}}{\sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1-1+n_2-1}}\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

where $\overline{x^{(i)}} = \frac{1}{n_i}\sum_{j=1}^{n_i} x_j^{(i)}$ and $s_i^2 = \frac{1}{n_i-1}\sum_{j=1}^{n_i}\left(x_j^{(i)} - \overline{x^{(i)}}\right)^2$, $i = 1, 2$

- Compare the p-value $\Pr[T < -|\tilde{t}| \text{ or } T > |\tilde{t}|]$ to significance level α, where $T \sim t_{n_1-1+n_2-1}$

Two-Sample Non-Pooled Variance t-Test

- If $\sigma_1 = \sigma_2$ is not assumed, we use a non-pooled variance test
- The test statistic is

$$\tilde{t} = \frac{\overline{x^{(1)}} - \overline{x^{(2)}}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- Compare the p-value $\Pr[T < -|\tilde{t}| \text{ or } T > |\tilde{t}|]$ to significance level α, where $T \sim t_{n_{\text{df}}}$, where

$$n_{\text{df}} = \left\lfloor \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2} \right\rfloor$$

- Note that $n_{\text{df}} \leq n_1 - 1 + n_2 - 1$ and the non-pooled variance test is less powerful than the pooled variance test

A TV Study Example: Matched Samples

A study aims to compare how long males and females spend watching TV. 20 married couples are randomly selected and the TV watching times for the husbands and wives are recorded.

- Consider $X^{(1)} - X^{(2)}$ for each couple
- The null hypothesis is

$$X^{(1)-X^{(2)}} \sim \mathcal{N}(0, \sigma^2)$$

- We can use a one-sample t-test to test it
- This usage of one-sample t-test is often called a paired t-test or a matched samples t-test
- It helps if $\sigma^2 < \sigma_1^2 + \sigma_2^2$